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Course: INSY 8311 & Database Development with PL/SQL

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Problem Statement

Pharmacies frequently face financial losses and patient safety risks due to poor stock management, unnoticed medicine expiration dates, and inaccurate demand forecasting. Manual monitoring leads to wasted drugs, critical stockouts, and accidental sales of expired medications.

Objectives

Main Objective: Build a normalized Oracle PL/SQL database to automate pharmacy inventory tracking, enforce drug safety rules, and predict future stock requirements.

Specific Objective

1. Design a 3rd Normal Form (3NF) relational database schema for suppliers, drugs, inventory batches, and sales.

2. Write PL/SQL triggers to automatically block the sale of expired drugs and alert on low stock levels.

3. Create PL/SQL procedures and packages for batch reception, sales processing, and inventory audits.

4. Develop an interactive dashboard to monitor expiry timelines and sales metrics in real time.

Data Source

Dataset: Pharma Sales and Drug Expiry Inventory Dataset (Kaggle)

Link: https://www.kaggle.com/datasets

Size: ~5,000+ records

License: Educational / Open Access

Planned Tables

1. SUPPLIERS: Holds medicine supplier contact and license details.

2. MEDICINES: Central drug catalog detailing brand names, categories, and safety storage guidelines.

3. STOCK_BATCHES: Tracks batch numbers, quantity received, manufacture dates, and expiry dates.

4. CUSTOMERS: Stores customer/patient details for prescription verification.

5. SALES: Transaction headers recording total billing, dates, and customer details.

6. SALE_ITEMS: Line items linking specific drug batches to sales transactions.

Planned PL/SQL Features

Function (check_stock_level): Returns current available stock for a specific medicine across all unexpired batches.

Procedure (process_drug_sale): Handles sales transactions and updates batch quantities automatically.

Trigger (prevent_expired_sale): A BEFORE INSERT trigger on SALE_ITEMS that blocks any attempt to sell a drug batch past its expiration date.

Package (pharmacy_inventory_pkg) Consolidates all inventory management, expiry alert routines, and batch updates into one encapsulated unit.

Innovation

Option: Analytics Dashboard

Tool: Power BI / Streamlit

Description: A live visual dashboard connected to Oracle database views displaying stock depletion trends, upcoming drug expiries (30/60/90 days), and high-demand medicine analytics.

References

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